Introduction to Molecular Dynamics

Laboratory for Chemistry and Life Science, Institute of Integrated Research, Institute of Science Tokyo
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1. Flow of MD simulation
2. Energy calculation
3. **Time integration**
4. Statistical mechanics; T & P control
5. Data analysis

Time integration: necessity and requirements

$$m_i \frac{\mathrm{d}^2 \mathbf{r}_i}{\mathrm{dt}^2} = -\frac{\partial U(\{\mathbf{r}_j\}_{j=1,2,\ldots N})}{\partial \mathbf{x}_i}$$

2nd-order differential eq.

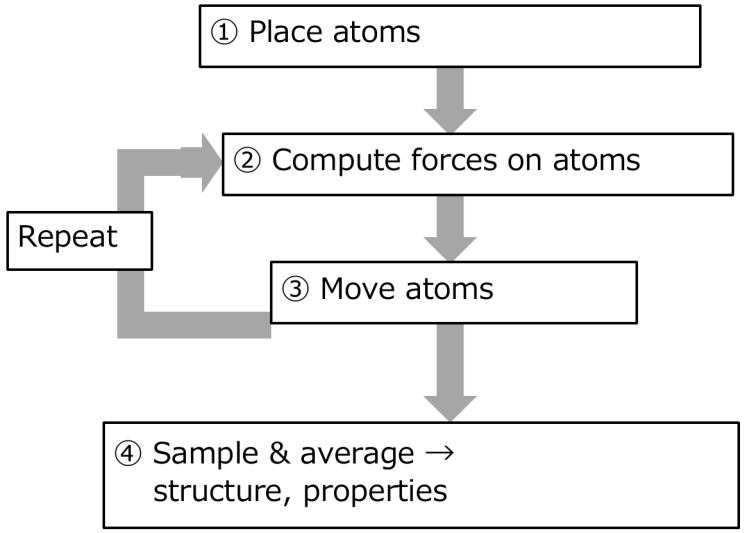
2 initial values: position, velocity

- Potential energy depends on 3D x*N* particle coordinates
- Generally no analytical solution → numerical integration is required

■ Requirements
- Energy conservation
- Time-reversal symmetry
- Low computational cost
  - Per Δt: only 1 energy calc.
- Long-term numerical stability

■ Common time integrators

(– Verlet) (– Predictor-corrector)
- **Velocity Verlet** (– Gear)
- Leap-frog



Velocity Verlet method

$$r(t+\Delta t) = r(t) + v(t)\Delta t + \frac{1}{2}\frac{F(t)}{m}\Delta t^2 + \mathcal{O}((\Delta t)^3)$$

$$v(t+\Delta t) = v(t) + \frac{\Delta t}{2}\left[\frac{F(t)}{m} + \frac{F(t+\Delta t)}{m}\right] + \mathcal{O}((\Delta t)^3)$$

- Algorithm
  1. Advance velocity by $\Delta t/2$
  2. Get position after Δt
  3. Acceleration at new positions (**compute forces**)
  4. Update velocity

$$v\left(t+\frac{\Delta t}{2}\right) \leftarrow v(t) + \frac{F(t)}{m}\frac{\Delta t}{2}$$

$$r\left(t+\Delta t\right) \leftarrow r(t) + v\left(t+\frac{\Delta t}{2}\right)\Delta t$$

$$v\left(t+\Delta t\right) \leftarrow v\left(t+\frac{\Delta t}{2}\right) + \frac{F(t+\Delta t)}{m}\frac{\Delta t}{2}$$

Choosing the time step

$t \rightarrow t + \Delta t \rightarrow t + 2\Delta t \rightarrow \ldots$

- Too small ⋯ correspondingly longer MD sampling
- Too large ⋯ numerically unstable, misses high momenta

System	Types of Motion	Time Step (s)
Atoms	Translation	10^{-14}
Rigid molecules	Translation and Rotation	$5x10^{-15}$
Flexible molecules, rigid bonds	Translation, Rotation and Torsion	$2x10^{-15}$
Flexible molecules, flexible bonds	Translation, Rotation, Torsion and Vibration	10^{-15} or $5x10^{-16}$

- Good choice ⋯ shortest timescale / 10-30 ≈ Δt

※ $1cm^{-1} = 3.0\times10^{10}$ Hz C-C: 1300 cm^{-1}, 25fs→ Δt = 1-2fs O-H: 3600 cm–1, 10fs→0.3-1fs

Keyword of advanced topic

Time evolution operator
- Liouville operator
- Suzuki-Trotter decomposition
- Symplectic
- Multi timestep (RESPA)

Constraint dynamics
- SHAKE
- RATTLE

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Ensembles

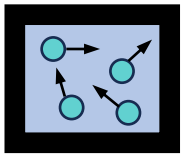
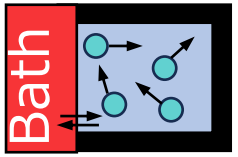
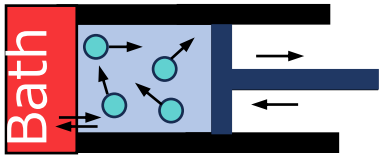
MD goal: **micro–macro connection**
- Interpret macro phenomena from micro
- Predict macro properties from micro eqs.

⇨ **Statistical mechanics** **+ Thermo-dynamics**

Ensemble Systems with different r, p but the same thermodynamic state

How to specify a system's "state"? (thermodynamics)
- Basically, fix 3 state variables

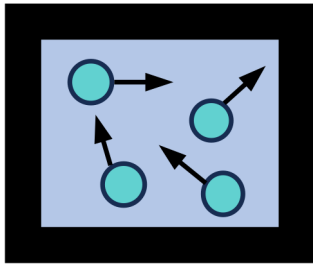
N ⋯ particles, V ⋯ volume, T ⋯ temperature,
E ⋯ internal energy, P ⋯ pressure

	Ensemble	Schematic
(1) *N*, *V*, *E*	Microcanonical ensemble (*NVE* ensemble)	
(2) *N*, *V*, *T*	Canonical ensemble (*NVT* ensemble)	
(3) *N*, *P*, *T*	*NPT* ensemble	

How to compute properties? How to link micro-macro?

Statistical mechanics: micro-macro connection

$$\begin{cases} \dfrac{\mathrm{d}\mathbf{r}_i}{\mathrm{d}t} = \dfrac{\mathbf{p}_i}{m_i} \\ \dfrac{\mathrm{d}\mathbf{p}_i}{\mathrm{d}t} = \mathbf{F}_i \end{cases}$$

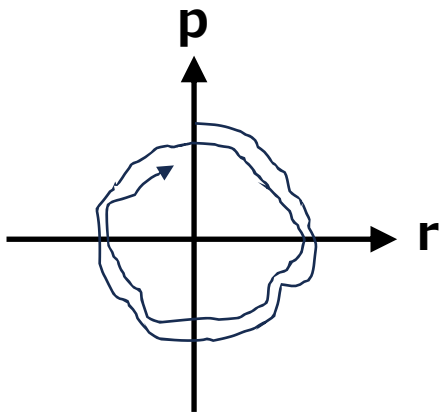


Microstate: phase-space point **Γ** = (**r**, **p**)

r = {**r**$_1$,**r**$_2$,...}

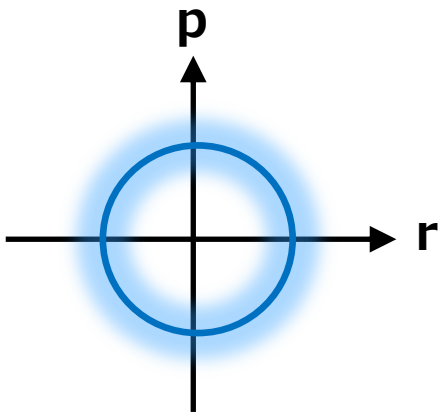
[Mechanics]
In principle, solving the equations of motion (left) for all trajectories tells us everything!
☹ **Generally intractable analytically**

Mechanics: trajectories



Γ = (**r**(t), **p**(t))

Stat. mech.: distribution



f(**r**, **p**)

[Statistical mechanics]
How is phase space **Γ** = (**r**, **p**) distributed?
: **distribution function** $f(\mathrm{r},\mathrm{p})$
→ gives macroscopic quantities!

Probability $\iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{p} f(\mathbf{r},\mathbf{p}) = 1$

Observable $\langle A \rangle = \iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{p} A(\mathbf{r},\boldsymbol{p}) f(\mathbf{r},\mathbf{p})$

(ensemble average)

Each ensemble has its own $f(\mathrm{r},\mathrm{p})$

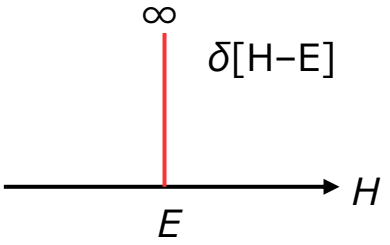
$f(\mathrm{r},\mathrm{p})$: how should we simulate to satisfy it?

Microcanonical ensemble (*NVE* ensemble)

N ⋯ *particles*
V ⋯ *volume*
E* ⋯ *internal energy

Distribution function

$$f_{NVE}(\mathbf{r},\mathbf{p}) = \frac{\delta[\mathcal{H}(\mathbf{r},\mathbf{p}) - E]}{\iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{p}\delta[\mathcal{H}(\mathbf{r},\mathbf{p}) - E]}$$



Energy is conserved
= plain Newton's equations ma = F

$$\mathcal{H}(\mathbf{r},\mathbf{p}) = \sum_i \frac{\mathbf{p}_i^2}{2m_i} + U(\mathbf{r})$$

Ensemble avg. $\langle A \rangle_{NVE} = \iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{p} A(\mathbf{r},\boldsymbol{p}) f_{NVE}(\mathbf{r},\mathbf{p})$

If the MD run explores **Γ**=(**r**, **p**) space widely enough⋯

Ensemble avg. = **Time average**

$$\langle A \rangle_{NVE} = \langle A \rangle_t = \lim_{\tau\to\infty} \frac{1}{\tau} \int_{t_0}^{t_0+\tau} A(t)\mathrm{d}t$$

Ergodic hypothesis

***NVE* is hard to compare with reality (e.g. hard to run MD at fixed T)** → ***NVT*, *NPT* ensembles needed**

Canonical ensemble (NVT ensemble)

N ⋯ *particles*
V ⋯ *volume*
T* ⋯ *temperature

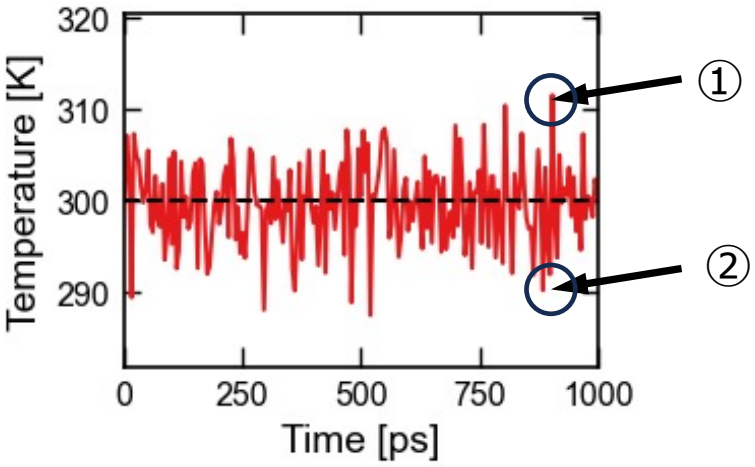
Distribution function

$$f_{NVT}(\mathbf{r},\mathbf{p}) = \frac{\exp[-\mathcal{H}(\mathbf{r},\mathbf{p})/(k_{\mathrm{B}}T)]}{\iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{p}\exp[-\mathcal{H}((\mathbf{r},\mathbf{p}))/(k_{\mathrm{B}}T)]}$$

Microscopic temperature

Equipartition law

$$\left\langle \sum_{i=1}^{N} \frac{\boldsymbol{p}_i^2}{2m_i} \right\rangle = \frac{3}{2}Nk_{\mathrm{B}}T \qquad T = \frac{1}{3Nk_{\mathrm{B}}} \left\langle \sum_{i=1}^{N} \frac{\boldsymbol{p}_i^2}{m_i} \right\rangle$$



Q. ① or ②: which is hotter?

Instantaneous temperature

$$\mathcal{T}(t) = \frac{1}{3Nk_{\mathrm{B}}} \sum_{i=1}^{N} \frac{\boldsymbol{p}_i(t)^2}{m_i}$$

$$\langle A \rangle_t = \lim_{\tau\to\infty} \frac{1}{\tau} \int_{t_0}^{t_0+\tau} A(t)\mathrm{d}t$$

Canonical ensemble (NVT ensemble)

N ⋯ *particles*
V ⋯ *volume*
T* ⋯ *temperature

Distribution function

$$f_{NVT}(\boldsymbol{r},\boldsymbol{p}) = \frac{\exp[-\mathcal{H}(\boldsymbol{r},\boldsymbol{p})/(k_{\mathrm{B}}T)]}{\iint \mathrm{d}\boldsymbol{r}\mathrm{d}\boldsymbol{p}\exp[-\mathcal{H}(\boldsymbol{r},\boldsymbol{p})/(k_{\mathrm{B}}T)]}$$

Temperature control methods

1. Berendsen — *Canonical ensemble*: **not generated**

2. Gauss constraint — *Canonical ensemble*:
not for momenta, but for positions

3. Nose-Hoover — ***Canonical ensemble*: generated**
Most common thermostat

4. Langevin — *Canonical ensemble*: generated
Total momentum not conserved,
so it must be removed at every step

Berendsen thermostat

N ⋯ *particles*
V ⋯ *volume*
T* ⋯ *temperature

Distribution function $f_{NVT}(\boldsymbol{r},\boldsymbol{p}) \neq \dfrac{\exp[-\mathcal{H}(\boldsymbol{r},\boldsymbol{p})/(k_{\mathrm{B}}T)]}{\iint \mathrm{d}\boldsymbol{r}\mathrm{d}\boldsymbol{p}\exp[-\mathcal{H}(\boldsymbol{r},\boldsymbol{p})/(k_{\mathrm{B}}T)]}$

- Canonical ensemble **not followed**
- Momenta scaled toward target T_0

$$\begin{aligned}\dot{\boldsymbol{r}}_i &= \frac{\boldsymbol{p}_i}{m_i} \\ \dot{\boldsymbol{p}}_i &= \boldsymbol{F}_i\end{aligned} \quad \Rightarrow \quad \begin{aligned}\dot{\boldsymbol{r}}_i &= \frac{\boldsymbol{p}_i}{m_i} \\ \dot{\boldsymbol{p}}_i &= \boldsymbol{F}_i + \frac{1}{\tau}\left(\frac{T_0}{\mathcal{T}(t)} - 1\right)\boldsymbol{p}_i\end{aligned}$$

Instantaneous temp.

$$\mathcal{T}(t) = \frac{1}{gk_{\mathrm{B}}}\sum_{i=1}^{N}\frac{\boldsymbol{p}_i^2}{m_i}$$

- More stable than Nose-Hoover
  → sometimes used for initial relaxation
  (followed by ..... a run satisfying canonical ensemble)

Nose-Hoover thermostat

N ⋯ *particles*
V ⋯ *volume*
T* ⋯ *temperature

Distribution function

$$f_{NVT}(\boldsymbol{r},\boldsymbol{p}) = \frac{\exp[-\mathcal{H}(\boldsymbol{r},\boldsymbol{p})/(k_{\mathrm{B}}T)]}{\iint \mathrm{d}\boldsymbol{r}\mathrm{d}\boldsymbol{p}\exp[-\mathcal{H}(\boldsymbol{r},\boldsymbol{p})/(k_{\mathrm{B}}T)]}$$

- **Follows canonical ensemble**
- Adds a thermostat degree of freedom ζ

$$\begin{aligned}\dot{\boldsymbol{r}}_i &= \frac{\boldsymbol{p}_i}{m_i}\\ \dot{\boldsymbol{p}}_i &= \boldsymbol{F}_i\end{aligned} \quad \Rightarrow \quad \begin{aligned}\dot{\boldsymbol{r}}_i &= \frac{\boldsymbol{p}_i}{m_i}\\ \dot{\boldsymbol{p}}_i &= \boldsymbol{F}_i - \zeta\boldsymbol{p}_i\\ \dot{\zeta} &= \frac{3Nk_{\mathrm{B}}}{Q}\left(\mathcal{T}(t) - T_0\right)\end{aligned}$$

$$\left(\Leftrightarrow \dot{\zeta} = \frac{1}{\tau^2}\left(\frac{\mathcal{T}(t)}{T_0} - 1\right)\right)$$

- Q: Fictitious mass
- ζ: friction coefficient

$$\mathcal{T}(t) = \frac{1}{gk_{\mathrm{B}}}\sum_{i=1}^{N}\frac{\boldsymbol{p}_i^2}{m_i}$$

- Inst. temp. $T(t)$ > target $T_0 \rightarrow \zeta \uparrow \rightarrow p \downarrow$
- Fictitious mass Q (or τ) must be set appropriately
  - Too large ⋯ weak coupling → slow temperature control
  - Too small ⋯ strong coupling → T oscillates, slow convergence

Pressure control

N ⋯ *particles*
P* ⋯ *pressure
T* ⋯ *temperature

Distribution function

$$f_{NPT}(\boldsymbol{r},\boldsymbol{p},V)=\frac{\exp[-(\mathcal{H}(\boldsymbol{r},\boldsymbol{p},V)+PV)/(k_{\mathrm{B}}T)]}{\iint \mathrm{d}\boldsymbol{r}\mathrm{d}\boldsymbol{p}\exp[-(\mathcal{H}(\boldsymbol{r},\boldsymbol{p},V)+PV)/(k_{\mathrm{B}}T)]}$$

Microscopic pressure

$$P=\frac{1}{3V}\left(\left\langle\sum_i^N\frac{\boldsymbol{p}_i^2}{m_i}\right\rangle+\left\langle\sum_{i<j}^N r_{ij}F_{ij}\right\rangle\right)$$

Barostat methods

1. Berendsen — *NpT* ensemble: not generated
2. Andersen — *NpT* ensemble: generated
3. **Parrinello-Rahman** — ***NpT* ensemble: generated**
   + allows cell deformation

Keyword of advanced topic

Nose-Hoover

- Nose-Hoover “chain” method
- Kinetic temperature, Configuration temperature
- Ergodicity of Nose-Hoover
  - Move in phase space based on the Boltzmann distribution
    ≠ Visit all points in phase space
    Ergodicity breaks down for harmonic oscillator
- Problem of scaling “Global” temperature
  - Multi-phase system, Imbalance of temperature distribution

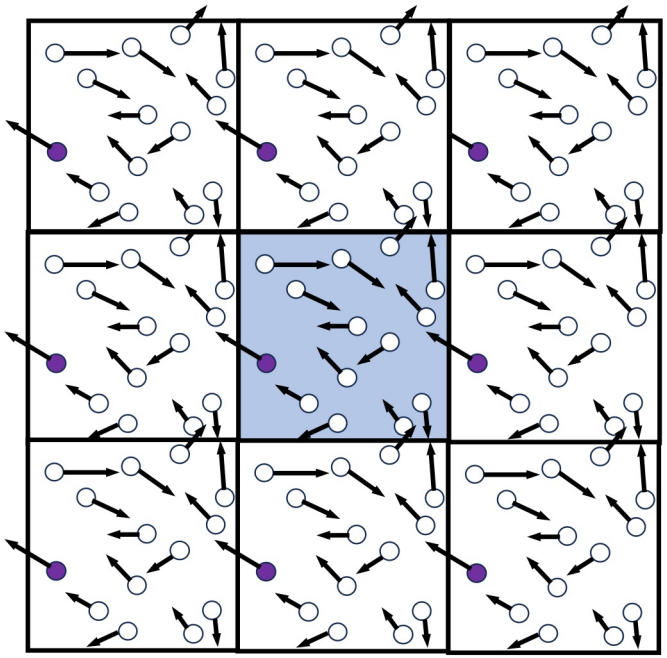
Pressure Control

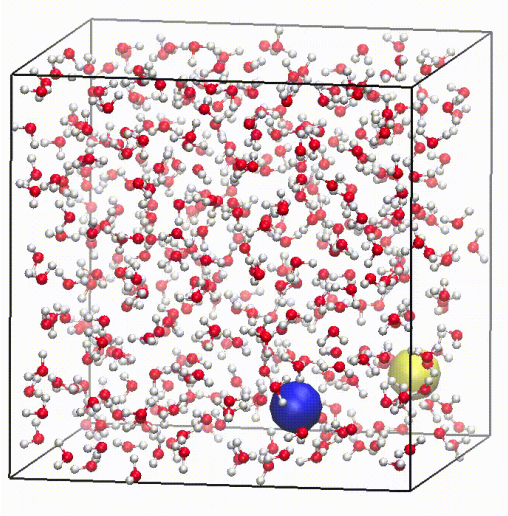
- Difficulty of calculating ⟨P⟩ / fluctuations too large…

Periodic boundary conditions

Treating cell edges as walls causes wall-interface effects
→ cannot be treated as bulk

Periodic BC　Upon exiting, ● re-enters from the opposite side





: cell explicitly simulated

Even a finite system (~10^2, 10^3,…) can be treated as bulk

Initialization & Equilibration

Modeling

↓

Energy minimization

↓

NVT ensemble
(Const. *N*umber of particles, *V*olume, *T*emperature)

Set temperature

↓

NpT ensemble
(Const. *N*umber of particles, *P*ressure, *T*emperature)

Equilibrium density

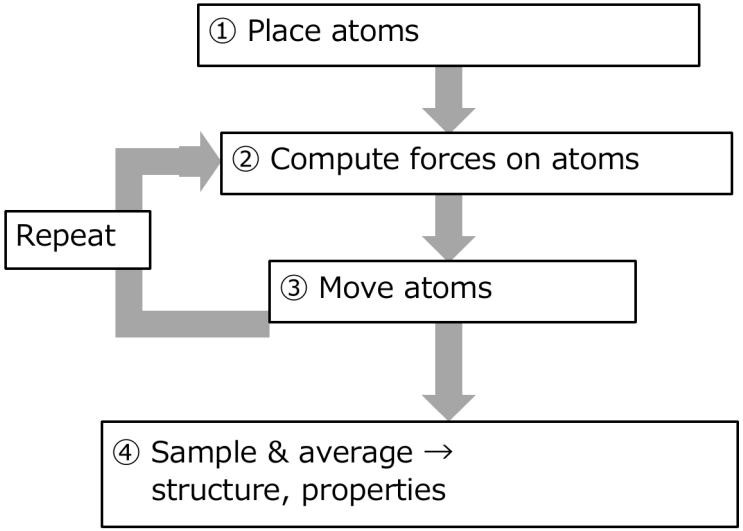
↓

Production run

Rarely use *NpT* for Production run
Usually *NVT* at equil. density

NpT: take great care with diffusion coefficients
Hummer *et al.*, *JCTC*, **19**, 3406 (2023).

Aside: Japanese contributions to MD simulation



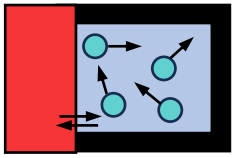
◎**Suzuki**-Trotter decomposition Masuo Suzuki
Velocity-Verlet × thermostat/barostat etc.
→ generalized time integration

$$e^{D_{\mathcal{H}}\Delta t} = e^{D_U \frac{\Delta t}{2}} e^{D_K \Delta t} e^{D_U \frac{\Delta t}{2}} + O((\Delta t)^3)$$

M. Suzuki, *Comm. Math. Phys.*, **51**, 183 (1976).

◎**Nosé**-Hoover thermostat Shuichi Nosé
Thermostat giving the canonical ensemble

$$\mathcal{H}_{\mathrm{N}} = \sum_{i=1}^{N} \frac{\boldsymbol{p}_i'^2}{2m_i s^2} + U(\boldsymbol{r}') + \frac{P_s^2}{2Q} + g k_{\mathrm{B}} T_0 \log s$$



S. Nosé, *J. Chem. Phys.*, **81**, 511 (1984). **2024: 40 years**

◎Green-**Kubo** formula Ryogo Kubo
Gives transport coefficients (diffusion, electrical & thermal conductivity, ⋯)

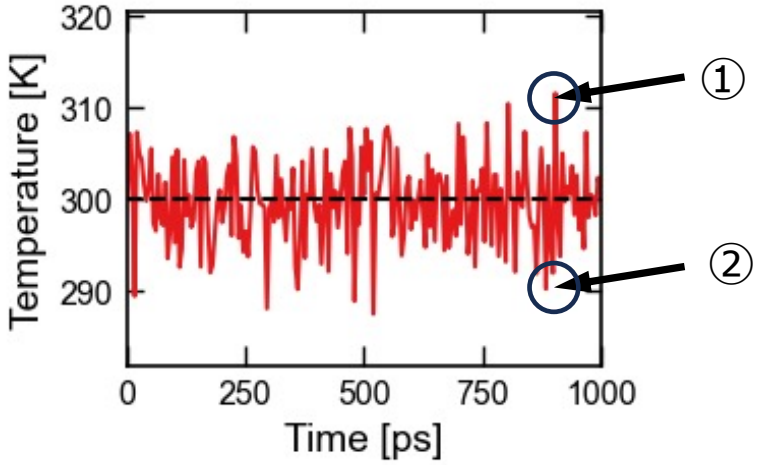
$$\gamma = \int_0^\infty \left\langle \dot{A}(t)\dot{A}(0) \right\rangle \, \mathrm{d}t.$$

R. Kubo, *J. Phys. Soc. Jpn.*, **12**, 570 (1957).

Assignment

Q1. Show that the Nosé thermostat generates the canonical ensemble.

Q2. An MD run gave the temperature vs. time shown below.
Discuss which is hotter, ① or ②.



Q3. “Accuracy” in MD includes energy accuracy and statistical accuracy.
Discuss the difference.